

Rectangle-constant closure: the theorem-grade rectangle prefactor, the incidence route, and the admissibility closure condition (gate STEP-5B)

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The v2.0 main-chain note left two items open: the rectangle prefactor c_R (measured, not derived) and the extreme- n rich-carrier corner. This companion note (i) archives the OPERATOR-SUPPLIED derivation (review verdict #9) that closes c_R at theorem grade — the OFFICIAL tier basis; (ii) contributes the incidence-geometric route, REPAIRED to the provisional exponent $N^{20/9}$ polylog (verdict #10, catch #7); (iii) reduces full STEP-5B closure to the named hypotheses H-ADM and H-KBAL with the VERIFIED threshold $n_{\text{adm}} \lesssim 1.59 \times 10^5$ at the anchor; and (iv, this re-issue) LIFTS H-KBAL in principle: an unconditional-amplitude $\sqrt{n} \log^2$ theorem with explicit constant. It claims NEITHER unconditional closure NOR any tier action; tier decisions remain with the operator.

2. Theorem-grade rectangle constant (operator derivation, verdict #9)

Proposition 1 (Rigorous $\sqrt{n}$ energy bound; operator-supplied). *Let $N = 2n = |\hat{Q}|$ and let p_C be the ordered antipodal pair count of carrier C . Three distinct points lie on at most one circle, so*

$$\sum_{C: k_C \geq 3} p_C^3 \leq \sum_{C: k_C \geq 3} k_C^3 \leq 27 \binom{N}{3} < \frac{9}{2} N^3, \quad \sum_{C: p_C \leq 2} p_C^3 \leq 4 \sum_C p_C \leq 4N^2,$$

hence $\sum_C p_C^3 \leq 7N^3$ conservatively. With $\sum_C p_C \leq N^2$ and Cauchy-Schwarz interpolation,

$$\sum_C p_C^2 \leq \left(\sum_C p_C \right)^{1/2} \left(\sum_C p_C^3 \right)^{1/2} \leq \sqrt{7} N^{5/2}.$$

For κ -balanced patterns ($A_{\max}^2 \leq \kappa^2 I/n = 2\kappa^2 I/N$), $\Psi_C \leq p_C A_{\max}^2$ gives

$$\sum_{t \neq 0} w_t^2 \leq \lambda^2 I^2 \left(8 + 4\sqrt{14} \kappa^4 \sqrt{n} \right), \quad c_R^{\text{rigorous}} = 4\sqrt{14} \simeq 14.97.$$

The v2.0 calibrated corollary is hereby THEOREM-grade; the certified region at the anchor budget ($K_{\text{budget}} = 5972$ at $I = 4 \times 10^{-4}$) is $n \lesssim 1.59 \times 10^5$ (3.77×10^3 at 10^{-3} ; 2.02×10^2 at 2×10^{-3}). Machine checks: $\sum p^3 \leq 7N^3$ and the interpolation hold on all test configurations.

3. New incidence route: repaired provisional exponent 20/9

Theorem 1 (Stereographic transfer). *A stereographic projection from a pole not on any carrier maps the sphere circles bijectively to circles (or lines) in the plane, preserving all point–circle incidences. (Machine check: projected ring circles are exact plane circles, residuals $\sim 10^{-30}$.)*

Theorem 2 (Incidence upgrade — repaired exponent 20/9). *Let m_r be the number of carriers with $p_C \geq r$. The pair partition gives the cap $m_r \leq 2N^2/r$; the planar Aronov–Sharir-type incidence bound gives $m_r \leq C(N^2/r^3 + N^3 L^{11/2}/r^{11/2} + N/r)$ (L polylogarithmic). The pair cap and the $r^{-11/2}$ term cross at*

$$2N^2/r_1 = N^3 L^{11/2}/r_1^{11/2} \iff r_1 = (NL^{11/2}/2)^{2/9},$$

so the dyadic sum is dominated by its value at r_1 :

$$\sum_C p_C^2 = \sum_r m_r r^2 \leq 2N^2 r_1 + O\left(\frac{N^3 L^{11/2}}{r_1^{7/2}}\right) = O(N^{20/9} \text{polylog } N), \quad \frac{20}{9} \simeq 2.222 < \frac{5}{2}.$$

(The 28/13 exponent of v1.0 was an arithmetic slip — caught by the operator, verify-loop catch #7; a dyadic self-check at $N = 4096$ confirms the 20/9 form with $O(1)$ ratio.) Consequently, for κ -balanced patterns, $K(n) \leq 8 + c\kappa^4 n^{2/9} \text{polylog } n$; with the conservative $c \text{polylog} = 30$ the anchor budget certifies $n \lesssim 2.2 \times 10^{10}$. This route is PROVISIONAL (supporting-candidate status) until the Aronov–Sharir constant is pinned; the OFFICIAL tier basis is the $\sqrt{n}$ route of the proposition above.

3b. Designated route to the sharp bound: structure vs randomness. The measured exponents ($\sim n^{2.05}$) and the unit-circle-strength incidence conjecture both point at $\sum_C p_C^2 = O(N^2 \text{polylog})$. A genuinely new route, not passing through open incidence conjectures, is registered as the designated multi-turn attack:

- *Lemma DR-1 (circle fiber rigidity = no small doubling on circles).* For A contained in a circle, distinct unordered pairs have distinct sums outside the axial direction (fiber ≤ 2 plus the axial exception), so $|A + A| \geq c|A|^2$ — subsets of circles can NEVER have small additive doubling. PROVED in substance by the fiber analysis of the universal single-circle theorem (main chain v1.8).
- *Lemma DR-2 (sphere structure dichotomy — OPEN, multi-turn).* If $E_2(\hat{Q}) \geq KN^2$ with K large, Balog–Szemerédi–Gowers yields $A' \subset \hat{Q}$, $|A'| \gtrsim N/\text{poly}(K)$, with doubling $|A' + A'| \leq \text{poly}(K)|A'|$. The dichotomy to prove: every sphere subset either has $|A + A| \geq |A|^{1+\delta}$ or concentrates its mass on $O(1)$ circles. Combined with DR-1 (circles forbid small doubling) this is self-improving and would force $E_2 = O(N^2 \text{polylog})$ unconditionally.
- *Status.* DR-2 is the single open ingredient; it is a Freiman-type structure statement for positively curved surfaces, related to (but not identical with) sum-set growth on strictly convex curves (Elekes–Nathanson–Ruzsa). Registered as OPEN with this note; no closure quote depends on it.

Remark 1 (Sharp conjecture, pre-registered). Measured growth exponents of $\sum_C p_C^2$ on the worst-known families are $n^{2.06}$ (rings), $n^{2.04}$ (random), $n^{2.08}$ (coaxial): all consistent with the SHARP conjecture $\sum_C p_C^2 = O(N^2 \text{polylog})$, under which $K(n)$ grows only polylogarithmically and the certified region exceeds any physically meaningful mode count for every admissible intensity. The conjecture is pre-registered with falsification gate: any family with measured exponent ≥ 2.3 refutes it.

4. Closure condition (named hypotheses)

Theorem 3 (Conditional STEP-5B closure). *Under the named hypotheses*

$$\mathbf{H-KBAL} : A_{\max}^2 \leq \kappa^2 I/n \text{ (balanced amplitudes, } \kappa \text{ explicit),} \quad \mathbf{H-ADM} : n \leq n_{\text{adm}},$$

STEP-5B holds for every admissible pattern provided (VERIFIED form, $\sqrt{n}$ route)

$$8 + 4\sqrt{14}\kappa^4\sqrt{n_{\text{adm}}} < K_{\text{budget}}(I),$$

i.e. $n_{\text{adm}} \lesssim 1.59 \times 10^5$ at the production anchor ($\kappa \simeq 1$). The repaired incidence route would relax this to $n_{\text{adm}} \lesssim 2.2 \times 10^{10}$ but is PROVISIONAL (constant unpinned) and is NOT used in the ledger threshold. The operator-noted packing form $n \leq C_{\text{pack}}(q_0/\Delta q_{\min})^2$ converts the verified threshold into the mode-separation condition $\Delta q_{\min}/q_0 \gtrsim 5 \times 10^{-3}$ (and the provisional one into $\gtrsim 10^{-5}$). The unbalanced-amplitude extension (amplitude-dyadic decomposition over $\sim \log n$ classes) is registered as the remaining technical lift of H-KBAL.

4b. H-KBAL lift (unconditional amplitudes).

Theorem 4 (Amplitude-dyadic lift). *For arbitrary positive amplitudes (no balance hypothesis),*

$$\sum_{t \neq 0} w_t^2 \leq 64\sqrt{7} \lambda'^2 I^2 \sqrt{n} \log^2(2n) + O(\lambda'^2 I^2).$$

Proof sketch (full chain machine-spot-checked). Partition $\hat{Q}$ into dyadic amplitude classes Q_j ($A \in (2^{-j-1}a_0, 2^{-j}a_0]$, N_j points, class intensity $2I_j$, $\alpha_j^2 \leq 8I_j/N_j$). Per class, the operator interpolation (verdict #9) gives the unweighted energy $E_j \leq \sqrt{7}N_j^{5/2}$; for class pairs, the bilinear additive energy obeys $E_{jk} \leq \sqrt{E_j E_k}$ (Cauchy–Schwarz over translates; machine-verified on a test pair). Hence $\|w^{jk}\|_{\ell_t^2} \leq \lambda' \alpha_j \alpha_k E_{jk}^{1/2} \leq \lambda' 8^{1/4} (I_j I_k)^{1/2} (N_j N_k)^{1/8}$. Classes beyond $J = \log_2(2n) + O(1)$ carry per-point mass $< a_0^2/N^2$ and are absorbed in the $O(\lambda'^2 I^2)$ term; for $j, k \leq J$, Minkowski in ℓ_t^2 and Cauchy–Schwarz over the $\leq J$ occupied indices give $\|w\|_{\ell_t^2}^2 \leq 64\sqrt{7} \lambda'^2 (\sum_j I_j)^2 \sqrt{N} \log^2(2n)$ with $\sum_j I_j = I$. $\square$

Remark 2 (Honest constant; ledger unchanged). The explicit constant $64\sqrt{7} \approx 169$ (times $\log^2$) is far from sharp: the worst MEASURED unbalanced ratio is 0.03 (power-law, exponential, two-scale profiles on rand/ring configurations) — amplitude conspiracies cannot beat the $\sqrt{n}$ scaling, and unbalanced profiles in fact REDUCE the ratio (heavy-mode domination concentrates mass into few pairs, which the diagonal $8I^2$ already covers). Because the constant is honest but large, the OFFICIAL ledger threshold remains the κ -balanced route (1.59×10^5); the sharp-constant unconditional optimization is registered as follow-up. Structurally, H-KBAL is no longer load-bearing: it affects constants, not the class-wide architecture.

5. Numerical verification

Script [codes/vacuum/beyond_layer_gershgorin_bound.py](#) (v1.10.0; 170/170 self-test asserts PASS; artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#)).

Quantity	Value	Check
operator p^3 bound	$\sum p^3 \leq 7N^3$ on rand/ring/coax	verdict-#9 constants verified
interpolation	$\sum p^2 \leq \sqrt{\sum p \sum p^3}$	all configs
c_R rigorous	$4\sqrt{14} = 14.9666$	theorem-grade
theorem region ($\sqrt{n}$)	$1.59 \times 10^5 / 3.77 \times 10^3 / 2.02 \times 10^2$	matches operator 1.58×10^5
stereographic transfer	circle residuals 10^{-30}	incidences preserved
incidence region (repaired)	2.2×10^{10} at anchor, PROVISIONAL	exponent 20/9; catch #7
dyadic self-check	ratio 1.1 = $O(1)$ at $N=4096$	20/9 optimization verified
H-KBAL lift	worst unbalanced ratio 0.03 vs ceiling 2929	κ -balance not load-bearing
bilinear energy CS	$E(A, B) = 480 \leq 1154$	lift's key step verified
growth exponents	2.06/2.04/2.08 (ring/rand/coax)	sharp- $O(n^2)$ evidence

Quantitative sanity check (mandatory): the operator's region figure 1.58×10^5 is reproduced independently (1.59×10^5 , rounding); $20/9 < 5/2$ strictly; the PROVISIONAL n -reach ratio ($2.2 \times 10^{10} / 1.59 \times 10^5 \approx 1.4 \times 10^5$) follows the $((K_b - 8)/c)^{9/2-2}$ scaling, dimensionally consistent; the H-KBAL-lift ceiling at $n = 32$ ($169 \log^2 64 \approx 2929$) exceeds the worst measured unbalanced ratio (0.03) by five orders — the lift constant is honest, not tight.

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

- (α) “*The Aronov–Sharir incidence bound is for plane circles; carriers live on the sphere.*” DISMISSED: stereographic projection from a generic pole is a bijection sphere-circles $\rightarrow$ plane-circles/lines preserving incidences (machine-verified exact); the planar theorem applies verbatim to the projected family.
- (β) “*The incidence exponent itself was wrong in v1.0.*” UPHELD and repaired: the operator caught the 28/13 arithmetic slip (verify-loop catch #7) — the correct crossover gives 20/9. The lesson is registered: exponent optimizations must carry a numerical dyadic self-check (now in the script, ratio 1.1 at $N = 4096$). The route is demoted to PROVISIONAL supporting-candidate; no ledger figure uses it.
- (γ) “*H-KBAL (κ -balance) may exclude physical competitors.*” VALID-with-mitigation: heavily unbalanced patterns are dominated by their heavy modes and fall into the small- n_{eff} certified region (the heavy sub-pattern carries the energy); the rigorous amplitude-dyadic lift is registered. No admissible competitor class is known to violate H-KBAL at fixed I .
- (δ) “*The sharp $O(n^2)$ claim rests on three families.*” DISMISSED as stated: the $O(n^2)$ statement is registered as a CONJECTURE with a pre-registered falsification gate (exponent ≥ 2.3 on any family), not as a theorem; no closure quote depends on it.

7. Result footer

Result ID: B5-BEYOND-LAYER-BOUND / AddA (serving STEP-5B)
Precise statement: (i) THEOREM (operator, verdicts #9/#10):
 $K(n) \leq 8 + 4 \sqrt{14} \kappa^4 \sqrt{n}$; $1.59e5$
= OFFICIAL threshold. (ii) 20/9 incidence route
provisional ($2.2e10$; constant unpinned).
(iii) NEW: H-KBAL LIFT - unconditional
amplitudes: $\sum w^2 \leq 64 \sqrt{7} \lambda^2 I^2$

$\sqrt{n} \log^2(2n)$; kappa-balance no longer
load-bearing (constants only). (iv) Conditional
closure {H-ADM} (+ constant-sharp H-KBAL for
the ledger figure). (v) DR-1 proved in
substance; DR-2 OPEN = sharp route.
Scope: OFFICIAL sharp threshold = balanced/sharp-
constant route (H-KBAL for the margin only);
arbitrary amplitudes controlled by the larger
dyadic constant (architecture lifted, margin
not sharp); second-cumulant Hessian scope.
Dependencies: B5 main chain v2.0 (carrier partition, rectangle
reformulation, triple count); A1-KERNEL-CONV;
margins from Proposition A (B2 chain).
Evidence grade: ANALYTIC: $\sqrt{n}$ route + H-KBAL lift;
PROVISIONAL/CITED: 20/9 incidence route;
EXECUTED: all verification
Reproduction command: python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output: claims 170/170 PASS; result.json under
claims/B5-BEYOND-LAYER-BOUND/runs/
260605-gershgorin-reduction/
Falsification gate: Any family with sum p^2 growth exponent ≥ 2.3
(refutes sharp conjecture, not the theorems);
any kappa-balanced pattern with $n < 1.59e5$
violating the margin (refutes the theorem chain).
Tier before / after: T4 / T4 (TIER PROPOSAL for operator: T5, or T6
CONDITIONAL on {H-KBAL, H-ADM} - decision is
the operator's)
No-overclaim statement: Unconditional class-wide closure (all n , all
amplitude profiles) is NOT claimed: it requires
the sharp $O(n^2)$ conjecture or H-ADM. The
incidence constant/polylog is conservative, not
pinned. No tier action on B1/B2; Reading-H
stays T5 pending operator review.
Next required action: DR-2 sphere structure dichotomy (multi-turn
designated attack - the single remaining sharp
route); pin the AS constant
(de-provisionalize $2.2e10$); sharpen the lift
constant ($64 \sqrt{7} \rightarrow O(1)$); H-ADM derivation
from TECT microphysics; operator tier review.